

# Functions of many variables – 4

## Homogeneity, scaling and functional equations

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### Theoretical minimum.

**Theorem 1 (Euler).** For a  $C^1$  function on  $\mathbb{R}^n \setminus \{0\}$  the identity  $f(tx) = t^k f(x)$  for all  $t > 0$  is equivalent to

$$Ef := x_1 \frac{\partial f}{\partial x_1} + \cdots + x_n \frac{\partial f}{\partial x_n} = kf.$$

One direction is differentiation in  $t$  at  $t = 1$ ; the other is the observation that  $\varphi(t) = t^{-k} f(tx)$  has zero derivative. Homogeneous functions are exactly the eigenfunctions of the Euler operator  $E$ , and  $E$  is a derivation:  $E(fg) = fEg + gEf$ .

**Theorem 2 (separating the degrees).** Substituting  $tx$  and comparing powers of  $t$  separates any polynomial into homogeneous components, and a relation holding for all  $t$  must hold for each component separately. A positively homogeneous function of degree  $k$  is determined by its restriction to the unit sphere:  $f(x) = \|x\|^k f(x/\|x\|)$ ; in particular a continuous function of degree 0 which extends continuously to the origin is constant.

**Theorem 3 (Hadamard's lemma).** If  $f$  is  $C^k$  near 0 and  $f(0) = 0$ , then  $f(x) = \sum_i x_i g_i(x)$  with  $g_i$  of class  $C^{k-1}$ , namely  $g_i(x) = \int_0^1 \partial_i f(tx) dt$ ; if moreover  $\nabla f(0) = 0$ , then  $f(x) = \sum_{i,j} x_i x_j h_{ij}(x)$ . This is how one divides by a coordinate without losing smoothness.

**Fact 1 (the rank of a function of two variables).** A representation  $f(x, y) = \sum_{i=1}^k g_i(x) h_i(y)$  exists if and only if the family of functions  $\{y \mapsto f(x, y)\}$  spans a space of dimension at most  $k$ , that is, if and only if every  $(k+1) \times (k+1)$  matrix  $(f(x_i, y_j))$  is degenerate. Linear independence is therefore the only obstruction; compare Polynomials – 8, problem 3.

**Fact 2 (how a functional equation in several variables is attacked).**

Substitute special values and coincidences of the variables; iterate the equation and look for a convergent process; find the group generated by the substitutions the equation allows, and describe its orbits — the solution is constant on them; only at the very end use continuity, which upgrades a solution on a dense set to a solution everywhere.

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1. Prove Euler's identity in both directions. Deduce that a function positively homogeneous of degree 1 and differentiable at the origin is linear, and that the gradient of a function homogeneous of degree  $k$  is homogeneous of degree  $k - 1$ . Give an example of a function homogeneous of degree 1 on  $\mathbb{R}^2$  which is not linear.
2. (Mexmat, 2012) Let  $f: \mathbb{R}^n \rightarrow \mathbb{R}$  be a homogeneous polynomial and suppose  $f(U + a) \subset [-1, 1]$  for some  $a \in \mathbb{R}^n$ , where  $U$  is a ball centred at the origin.
  - a) Prove that  $f(c \cdot U) \subset [-1, 1]$  for  $c = \frac{1}{6}$ .
  - b) For which  $c > 0$  is the assertion valid independently of  $n, f, a$  and  $U$ ?
3. (Proposed problems) Prove that the only twice differentiable functions  $f: (0, \infty)^2 \rightarrow \mathbb{R}$  satisfying

$$x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} + \frac{1}{4}f = 0$$

are  $f(x, y) = C(x, y) + D(x, y) \ln x$ , where  $C$  and  $D$  are twice differentiable homogeneous functions of degree  $\frac{1}{2}$ . (The left-hand side is  $E(E - 1)f + \frac{1}{4}f = (E - \frac{1}{2})^2 f$ .)

4. (Mexmat, 2013) Given a natural number  $n$ , consider the function of two real variables  $f(x, y) = x^n + x^{n-1}y + x^{n-2}y^2 + \dots + xy^{n-1} + y^n$ . Find the minimal  $k$  for which there exist functions  $g_1, \dots, g_k, h_1, \dots, h_k$  of one real variable with  $f(x, y) = \sum_{i=1}^k g_i(x)h_i(y)$ .

5. (Высшая лига 2025) Does there exist a continuous non-constant function defined on  $\mathbb{R}^2 \setminus \{0\}$  such that for every integer matrix  $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$  with  $ad-bc = 1$  one has  $f(ax+by, cx+dy) = f(x, y)$  for all  $(x, y) \in \mathbb{R}^2 \setminus \{0\}$ ?